Rising vapor-pressure deficit increases nitrogen fixation in a legume crop

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Summary

- Atmospheric vapor-pressure deficit (VPD) is increasing in many regions and has a large impact on plant productivity. A VPD increase leads to raising transpiration rate (TR) and soil-water demand, risking productivity penalties. Like water, nitrogen is critical to productivity, but the effect of VPD on legume nitrogen fixation is undocumented.
- To address this, we developed a portable system for quantifying nitrogen fixation noninvasively and at a high temporal resolution by tracking the rate of hydrogen gas evolution by root nodules. Combining field and controlled-environment experiments where we measured leaf gas exchange and H_2 production by nodules, we confirmed the ability of the system to track nitrogen fixation dynamics.
- Raising VPD from 0.5 to 3 kPa within *c.* 2.5 h under well-watered conditions increased nitrogen fixation by up to 25% in addition to TR, consistent with the hypothesis that raising VPD in that range might have alleviated nitrogenase feedback inhibition. Genotypic differences were found in this response, indicating a potential for breeding.
- Our study provides evidence for an important environmental effect on nitrogen fixation that is not taken into account in current crop and vegetation models, pointing to untapped avenues for better understanding climate change effects on legumes and nitrogen cycling.

Introduction

Atmospheric drought or vapor-pressure deficit (VPD) is globally on the rise, negatively impacting plant productivity world-wide (Yuan *et al.*, 2019). For legumes such as soybean, variation in VPD is one of the major yield-limiting factors (Zhang *et al.*, 2017; Kimm *et al.*, 2020; Zhou *et al.*, 2020). For instance, in the United States, a meta-analysis conducted on soybean yield data assembled from 27 soybean-producing states during the 2007–2016 period found that VPD from 61 to 90 d after sowing was the most important environmental predictor of soybean yield across the country (Mourtzinis *et al.*, 2019). The mechanistic basis for this negative effect of rising VPD on field-grown soybean is understood to arise from its effect on increasing transpiration rate (TR) and the resulting risk of buildup of soil-water deficit during seed fill, based on well-established biophysical relationships (Tanner & Sinclair, 1983; Sinclair *et al.*, 2010). To counteract this effect, the existing diversity in soybean TR response curves to increasing VPD has been used to breed for cultivars that are more productive under high VPD, by exhibiting a water-saving behavior that makes more soil water available during seed fill (Sadok & Sinclair, 2010; Carter *et al.*, 2016; Ye *et al.*, 2020).

In addition to water, soybean production is critically dependent on the availability of nitrogen, mostly through the key process of biological nitrogen fixation, which is responsible for fixing up to 90% of nitrogen used by the crop (Ciampitti & Salvagiotti, 2018). Surprisingly however the effect of VPD on soybean nitrogen fixation has received considerably less attention and in fact, there is currently no direct evidence relating evaporative demand to nitrogen fixation on any nitrogen-fixing plant species. Recently, we (Lopez *et al.*, 2021) conducted a systematic review on the physiological basis of plant response to VPD, independent from any confounding variable, on 112 species and 56 traits, finding no evidence of this relationship being investigated over the last 50 yr. The closest evidence is available, stemming from environmental association studies leveraging seasonal weather data in the field, point to positive (Ciampitti *et al.*, 2021) or negative (de Borja Reis *et al.*, 2021) correlation between VPD and nitrogen fixation, likely due to VPD co-varying variables such as PAR, temperature, and soil moisture in such environments.

In the absence of such direct evidence, an assumption would be that VPD increase would negatively regulate nitrogen fixation by triggering soil-moisture deficits. This is based on long-standing evidence showing that nitrogen fixation decreases in response to progressive soil drying earlier than most other physiological processes, including photosynthesis (Durand *et al.*, 1987; Sall & Sinclair, 1991; Purcell *et al.*, 1997; Serraj & Sinclair, 1997). This sensitivity to soil-water deficit was hypothesized to be the result of a feedback-inhibition mechanism arising from lower TR caused by the buildup of soil-moisture deficit leading to the accumulation in nodules of fixed nitrogen in the form of ureides, which inhibits nitrogen fixation (Sinclair & Serraj, 1995; King & Purcell, 2005; Ladrera *et al*., 2007). Alternatively, this feedback inhibition may originate from the shoots, leading to the accumulation of ureides in the nodule via phloem flow (Parsons *et al*., 1993; Vadez *et al*., 2000). This mechanism was hypothesized to arise from an increase in xylem-to-phloem translocation of nitrogen compounds through transfer cells triggered by a decline in TR with water limitation (Da Silva & Shelp, 1990; White, 2012).

Based on the above evidence, our hypothesis is that under conditions where soil-moisture deficit is minimized or null, an increase in VPD may lead to an *enhanced* nitrogen fixation under well-watered conditions by alleviating both potential sources of feedback inhibition, through increasing xylem water flow due to increased TR. To test this hypothesis, we developed a new and cost-effective approach to nondestructively quantify nitrogen fixation in parallel with leaf gas exchange under field and controlled-environment conditions where VPD effects could be isolated and quantified.

Currently, the most widely used technique for monitoring nitrogen fixation rates nondestructively is the acetylene-reduction assay (ARA). This method measures the activity of dinitrogenase, which, in the absence of dinitrogen, will instead reduce acetylene to ethylene (Hardy *et al*., 1968). This technique, however, presents a number of drawbacks, including the requirement of specialized and expensive equipment, the highly flammable nature of acetylene gas (Parsons *et al*., 1992; Hunt & Layzell, 1993), and the potential for endogenously formed ethylene (Smercina *et al*., 2019). Additionally, exposure to saturating acetylene can trigger dramatic decreases in nitrogenase activity (Minchin *et al*., 1983; Silvester *et al*., 1989). An alternative approach is to measure hydrogen (H_2) gas, a by-product of nitrogen fixation, which avoids many of the problems with acetylene reduction (Hunt & Layzell, 1993). This method has been used successfully under controlled conditions (Witty & Minchin, 1998; Oono *et al*., 2020), but we designed a system which can also be used in the field.

The goal of this investigation was to examine rates of nitrogen fixation in two elite soybean genotypes, in relation to concurrent changes in environmental variables under field and controlled-environment conditions, with a focus on VPD effects at a high temporal resolution (*c*. 2.5 h), independent of any other confounding variables. Measurements of gas-exchange variables (transpiration, stomatal conductance, and photosynthesis) and root mass, nodule number, and nodule fresh mass were conducted in parallel with tracking nodule H_2 production to test the hypothesis that nitrogen fixation positively responds to increases in atmospheric VPD.

Materials and Methods

Plant material

Two soybean (*Glycine max* L.) genotypes were used in the study, consisting of MN1312CN and CZ1418LL, two elite high-yielding varieties. These genotypes were selected based on their ability to grow well in polyvinyl chloride (PVC) tubes placed in the field (see next section) in a preliminary trial that included an original group of 15 genotypes.

Field experiment

Planting and growth conditions Single-row soybean plots (3.7-m rows replicated six times per genotype) were machine-planted at the University of Minnesota field plots on the St Paul campus on 5 June 2018. Following this planting, 30-cm-long PVC tubes, with a 10-cm diameter were inserted into the center of each plot. These tubes are part of the setup designed for measuring nitrogen fixation (see the next section). Half of these were hammered directly into the soil and therefore contained native field soil, while the other half were placed into dug-out holes and filled with coarse sand. The sandy soil was used in case the native soil harbored carry-over nitrogen levels high enough to downregulate atmospheric nitrogen fixation or particular rhizobial strains which could have dampened the H_2 signal used for tracking nitrogen fixation (to be described later) by recycling it within the nodule.

Three seeds per genotype were manually planted in the center of the PVC opening at a depth of 2.5 cm, and plants were thinned to one plant per tube 14 d after germination. The day after planting, all tubes were inoculated with rhizobia suspended in nitrogen-free nutrient solution. The solution consisted of a mixture of two rhizobial strains native to local MN field soils. The inoculation was repeated a second time 1 wk later as seedlings emerged.

To supplement soil nutrients that were unavailable to the seedlings in sand tubes, a nitrogen-free nutrient solution was prepared based on Fujikake *et al.* (2002) and applied to sand tubes once per week. This nutrient solution was given to the sand tubes for 9 wk after planting until the roots were growing freely out of the bottom of the PVC tubes and directly accessing field soil. Plants grown in sand-filled tubes were watered every 2 d on average throughout the season to minimize the risk of soil-moisture deficit. From the date of planting through the end of the experiment, plants grew under conditions reported in Supporting Information Table S1 and Fig. S1.

***In situ* measurement of root system H_2 production rate** A portable system was custom-built to compare rates of nitrogen fixation by intact soybean plants. This device measures production of hydrogen gas (H_2) by the nitrogenase. Using this approach to estimate nitrogen fixation in real time was pioneered by Witty (1991) and Hunt & Layzell (1993) and further developed by Oono *et al.* (2020). The system uses an air pump connected to a H_2 sensor, both controlled by an Arduino circuit board and a software that enable the user to control settings for pump speed and sampling interval (set at 1 s). The device is versatile, making it possible to quantify H_2 levels on plants grown under field and growth-chamber conditions as illustrated in Fig. S2.

Under field conditions, H_2 from the nodulated roots was sampled from 'root chambers' following a design previously successfully used with the ARA (Denison *et al*., 1983a,b; Fig. S2), but with a longer PVC tube (30 cm) to sample more of the root system. When plants grown in the PVC tube reached the desired growth stage (R1–R3), 2-cm-thick lids made of rubber were sealed around the stem. A 2-mm hole was drilled through the lid to enable connection to the H_2 sensing equipment via a 1-m long silicone tube.

During measurements, the device was run from a laptop so that H_2 production rates could be monitored in real time. Measurements on the two selected genotypes began *c.* 2 months after planting, taking place over three consecutive weeks (stages R1–R3), every 4 d on average, under conditions reported in Table S2. This yielded six sampling days where two genotypes were screened for root H_2 production twice per day, in the morning (09:00–12:00 h) and afternoon (13:00–16:00 h). In each round of measurements, both sand and native-soil tubes were measured, for a total of 132 measurements (2 genotypes × 3 replications × 2 soil types). On one afternoon, the H_2 sampling was not conducted due to a device malfunction.

During measurements, the pump speed was set identically for each sampling day and the sampling interval was always set to 1 s, with each measurement lasting for a total of 270 s. Before assaying roots, the pump was run for 90 s to sample ambient air. At 90 s, the H_2 sensing equipment was rapidly connected to the lid-mounted gasket and the sampling then ran for 180 s. In nearly all cases, H_2 production dynamics followed a sigmoidal time course: apparent ambient air H_2 concentration reading ('zero' setting was slightly positive, because Arduino cannot read negative voltages) was at a steady state before sampling began, after which H_2 progressively increased to reach a new plateau (Fig. S3). The baseline apparent H_2 levels observed before pump attachment were subtracted to get actual H_2 production rate (see Data processing and statistical analysis).

H_2 production was attributed to nitrogen fixation by nodules in the soybean root system enclosed in the tube (Denison *et al.*, 1983a,b; Oono & Denison, 2010). Transformation of voltage readings into ml H_2 per plant h^{-1} was based on sensor calibration with a 50-ppm H_2 gas standard and measurements of flow rate (Witty & Minchin, 1998; Walch *et al.*, 2014; Oono *et al.*, 2020). As a result, the rate of H_2 production is given by this equation:

$$H_2 \text{ production rate} = H_2 \text{ concentration} \times \text{flow rate} \qquad \text{Eqn 1}$$

where the flow rates were estimated for both soil types using a flow meter.

Leaf gas-exchange measurements in the field Leaf gas-exchange data were recorded on two of the 6 d when H_2 production assays took place (i.e. on the same plants), using an LI-6800 Photosynthesis System (Li-Cor Inc., Lincoln, NE, USA) such that leaves covered 100% of the cuvette area. Measurements of stomatal conductance to water vapor (g_{sw}, mol H_2O m^{-2} s^{-1}), TR (E, mol H_2O m^{-2} s^{-1}), and photosynthetic assimilation rate (A, μmol CO_2 m^{-2} s^{-1}) were conducted on the middle section of the middle leaflet of the third trifoliate, twice per day, during mornings and afternoons. Cuvette conditions were set as follows: $[CO_2] = 400$ ppm (sample analyzer), flow rate = 500 μmol s^{-1}, fan speed = 10 000 rpm, and overpressure = 0.2 kPa. Temperature and relative humidity were set to track and match the environmental conditions, and the headlight source was set to track PAR sensor readings and match the ambient light levels. For each measurement, two stability thresholds had to be met to record the considered value. Sample-to-reference difference in CO_2 levels (ΔCO_2) slope had to be below 0.7 over a 20 s period. The second stability threshold was the slope of the sample-to-reference difference in water vapor levels (ΔH_2O), which value was set to 0.2 over a 20 s period.

Growth-chamber experiments

Plant preparation and growth conditions Four experiments (E2–E5) were conducted under controlled-environment conditions on the same two genotypes, where plants were subjected to four VPD treatments (low VPD: 0.5 and 0.8 kPa; and high VPD: 2.0 and 3.0 kPa) under constant PAR of *c.* 470 μmol m^{-2} s^{-1} and temperatures of 25°C (0.5 and 2.0 kPa) and 30°C (0.8 and 3.0 kPa). As detailed in Table S3, experiments were undertaken such that the same batch of plants were subjected to the low and high-VPD treatments sequentially, which is the typical progression of VPD occurring in the field from the morning to the middle afternoon (Sinclair *et al.*, 2017).

Plants were grown inside a walk-in growth chamber (Model PGV36; Conviron, Controlled Environments Ltd, Winnipeg, MB, Canada) with the following settings: daytime : nighttime temperature = 25°C : 18°C, daytime : nighttime VPD = 1.5 kPa : 0.5 kPa, PAR = 450 μmol m^{-2} s^{-1}, and photoperiod = 14 h. Plants were seeded in 50 × 15 cm plastic growth pouches which have a paper wick to retain moisture, hold seeds, and support plants after germination (Oono & Denison, 2010; Fig. S2). The bottom crease of this folded section is serrated with holes that allow roots to easily penetrate and grow into the large rooting area in the growth pouch below.

Before placement in the growth pouches, seeds were sprayed with a 70% ethanol solution, soaked in hydrogen peroxide for 2 min, and then rinsed in distilled water to ensure sterilization and improve germination. Two seeds were then placed hilum-down in each pouch to ensure successful germination, and each pouch was thinned to a single healthy plant 10 d after germination. Initially, each pouch received 50 ml of a nitrogen-free nutrient solution prepared following Fujikake *et al.* (2002). A volume of 1 ml of one rhizobial inoculant used in the field experiment (*c.* 2×10^6 rhizobia cells) was added to each pouch at seeding and second time when taproots were *c.* 3 cm long. Ten days after seeding, nodulated plants received a nitrogen-free nutrient solution, two to three times per week, with the amount of added solution calculated to replace average daily water loss.

In E4, some plants were not inoculated to confirm that plants without nodules do not emit hydrogen (see the Results section). This run included eight inoculated plants (4 per genotype, Table S3) watered with the nitrogen-free nutrient solution, and eight noninoculated plants supplied with the same nutrient solution with supplemental nitrogen provided by the addition of 10 mM ammonium nitrate. In some instances, these plants still presented a very small number of nodules, which were removed manually.

In all experiments, pouches were arranged in grid containers with dividers, which were inside the chamber rotated twice per week, while pouches were rotated once per week within the bin. Sensors were deployed in three locations across the growth chamber to record temperature, relative humidity, and VPD at the plant canopy height every 5 min (EL-USB-2-LCD; Lascar Electronics, Whiteparish, UK). Photosynthetically active radiation (PAR) was recorded via a quantum sensor (Hobo H21-USB micro station; Onset Computer Corp., Bourne, MA, USA) similarly placed at canopy height in the center of the growth chamber.

Nitrogen fixation response to VPD The effect of VPD on nitrogen fixation was investigated at two temperatures (25°C, in E2 and E4; and 30°C, in E3 and E5). At each temperature, the low- and high-VPD treatments were imposed based on protocols from Tamang & Sadok (2018) and Tamang *et al.* (2022). Briefly, at each temperature and the low-VPD treatment was achieved using four industrial humidifiers which made it possible to maintain a relative humidity (RH) above 80%. The high-VPD treatment was imposed after turning off the foggers and activating two industrial dehumidifiers, which were programmed to maintain a RH of 30%.

In each experiment, the sequence started with the low VPD treatment initiated at the beginning of the light period (06:00 h), to which plants were acclimated for 60 min. Afterward, nitrogen fixation measurements were conducted under these same conditions for an additional 90–120 min. After these measurements were completed, the same plants were then acclimated to the high-VPD treatment for the same duration (60 min) after which the high-VPD hydrogen measurements were conducted during the subsequent 90–120 min following the same plant-sampling order as in the low VPD treatment. Experimental conditions of the controlled-environment measurements are compiled in Table S3.

Comparisons of nitrogen fixation rates were conducted by tracking rates of H_2 production by the nodulated roots. This was done using a similar portable H_2 sensing device as the one used in the field experiment, and following the same general approach described earlier, with some modification due to the nature of the controlled-environment plant growth setup (Fig. S2). Briefly, before measurements, the pouches were drained of nutrient solution so that the only moisture remaining was the amount of water adsorbed on root and pouch wall (Noel *et al.*, 1982). This approach was needed to ensure consistent and adequate aeration during each pouch measurement and ensure that no nodules were submerged underwater (Serraj & Sinclair, 1996). The H_2 sampling device was connected to the pouch via male-to-female Luer lock fittings attached to a metal tube feeding through a rubber gasket which fits tightly into the top of the growth pouches (Fig. S2).

Similar to the field measurements, the pump speed was set identically for each sampling day and H_2 levels were recorded every second. Ambient air was sampled for 60 s before connecting the instrument to the pouch gasket via Teflon capillary tubing, in order to quantify a baseline H_2 detection. Measurements were then conducted for 500 s before the process is repeated with the next plant. The longer time for sampling in the controlled environment compared with the field is based on pilot tests for how long it would take for H_2 to reach a steady state in the growth pouch (data not shown). After the pump hose was detached, the gasket was removed from the pouch and sterilized with isopropyl alcohol to prevent the spread of rhizobia between pouches, and nutrient solution was resupplied. Relative H_2 units were transformed into absolute values (ml H_2 plant^{-1} h^{-1}) as described earlier with a new measurement of flow rate and calibration with a 50 ppm H_2 standard.

To ensure that the influence of VPD on nitrogen fixation was studied independent from potential endogenous (e.g. ultradian or circadian) effects, two additional experiments were conducted on both genotypes to test whether the time of measurements had an effect on the nitrogen fixation rate in two independent experiments at 25°C (12 plants) and 30°C (6 plants), on plants that were 48–49 d old. At 25°C, plants were subjected to a constant VPD treatment of 1.3 kPa with nitrogen fixation assayed at 07:35 and 12:15 h, while at 30°C, plants were exposed to constant VPD of 1 kPa, with nitrogen fixation measured at 07:40 and 12:25 h. Both experiments were conducted under a constant PAR of 440–450 μmol m^{-2} s^{-1}.

Leaf gas-exchange response to VPD Leaf gas-exchange variables (A, g_{sw}, E) were measured in E5 (E5.1, E5.2, and E5.3, Table S3) under temperature conditions that maximized VPD effect on nitrogen fixation (30°C, see the Results section). Similar to field conditions, measurements were undertaken on the middle leaflet of the third trifoliate, on plants that were 46–69 d old. These were carried out over 2 d; each time on the day immediately following the one where H_2 sampling took place. As was the case with the H_2 sampling, measurements under target VPD conditions were conducted after acclimating plants to the low (0.8 kPa) and high (3 kPa) VPD conditions for 60 min. The LI-6800 settings for T and PAR cuvette conditions were set to the ambient conditions (T = 30°C, PAR = 450 μmol m^{-2} s^{-1},) and RH was set at 84% and 29% for the low and high-VPD steps, respectively. The rest of the settings were identical to the field experiment except that overpressure was set at 0.1 kPa.

Plant size and nodule measurements Plant phenological stages, total leaf areas, shoot and root dry mass, and nodule traits were determined for all plants in experiments E2, E3, E4, and E5. Nodule fresh mass was estimated based on root system images taken using an iPhone 6s camera (Apple Inc., Cupertino, CA, USA). Based on an assumption of equivalent nodule density and spherical shape, fresh weights were estimated from the nodule volume, which was determined based on nodule diameter as calculated by the RootPainter program, a GUI-based software program to rapidly train deep neural networks for use in biological image analysis (Smith *et al.*, 2020), trained on our root images. Dry mass values were recorded by weighing plant material using a balance after oven-drying for 3 d at 60°C.

Data processing and statistical analysis

For field-based H_2 time courses, the first 45 s of the 90 s ambient air sampling was excluded from the analysis as H_2 readings were occasionally still decreasing from the previous plant sample in the first 30–45 s of the following measurement. Time courses of H_2 production were then fitted with a sigmoidal curve which enabled computing a baseline value for ambient air H_2 (measured before connecting the device to the root chamber), and a higher plateau, reflecting steady-state H_2 production by the nodules (Fig. S3). For each measurement, the actual rate of H_2 production was estimated as the value of the high plateau minus the baseline value. This ensured that variation in the background H_2 levels is taken into account to accurately compute H_2 values that reflect nitrogen fixation by the nodules.

In the case of growth-chamber data, a one-phase exponential-decay model was found to be a better fit for the H_2 time courses after sampling starts. This was primarily due to the presence of a H_2 peak that manifests right as the pump is connected to the pouch gasket due to H_2 buildup in the small pouch volume. The H_2 levels then progressively decrease to a steady-state value, representative of the H_2 production rate by the nodules (Fig. S3). An average H_2 baseline value was calculated before each measurement, based on the 30 s of ambient air sampling immediately before pump-pouch attachment, and subtracted from each steady-state H_2 level derived from the one-phase, exponential-decay model.

Differences between genotypes and treatments were analyzed using parametric *t*-tests and two-way ANOVAs. Parametric *t*-tests, regression analyses, and Pearson correlations were conducted using the statistical software PRISM v.7.0c (2017; GraphPad Software Inc., San Diego, CA, USA). Analyses of variance combined with Tuckey's range tests were conducted using R-based scripts (R Core Team, 2021). Relative values for gas exchange and H_2 production were calculated as ratios of the high-VPD to the low-VPD measurements.

Results

Validation of the assay: effect of nodule presence, sampling time, and genotype on H_2 production

The controlled-environment setup was used to test whether H_2 production could be traced to nodule presence and that it was not affected by uncontrolled endogenous factors. As shown in Fig. 1, H_2 production for plants without nodules was not significantly different from zero for both genotypes. In sharp contrast, nodulated replications of the examined genotypes, which were examined under the same temperature, VPD, and PAR conditions as their denodulated counterparts, emitted substantial levels of H_2 gas, indicative of nitrogen fixation activity arising from nodule presence. Additionally, when measured under constant temperature, VPD, and PAR, H_2 levels were not found to be significantly influenced by the timing of the sampling for both genotypes, whether nitrogen fixation was assayed at 25°C or 30°C ($P = 0.08$–0.92 depending on the genotype and the temperature treatment, Fig. S4).

Genotype CZ1418LL consistently exhibited higher rates of nitrogen fixation compared with MN1312CN (Fig. 2). Interestingly, genotypic differences in the rate of H_2 production were highest when expressed on the basis of nodule number (Fig. 2c, 70% higher for CZ1418LL) and minimized when normalized based solely on root dry mass (Fig. 2b, 5% higher for CZ1418LL). In the latter case, removing a single outlier datum for CZ1418LL (indicated by an arrow in Fig. 2b) yielded non-significant genotypic differences, indicating that differences in nodule-specific activity (ml H_2 mg^{-1} nodule h^{-1}), rather than nodule mass per plant, were key to genotypic differences in H_2 production.

Nitrogen fixation and gas-exchange response to fluctuating environment in the field

When deployed under field conditions, the H_2 sensing system consistently showed that independent from genotype and combining data from both soil types, H_2 production increased significantly ($P < 0.0001$) by 46% in the afternoon across all dates (Fig. 3). Both genotypes exhibited the same pattern (Fig. 3), concomitant with increased temperature, PAR, and VPD (Table S1; Fig. S1). This pattern was particularly strong in plants grown in sandy soil, where root H_2 production significantly increased by 60% and 44% for genotypes MN1312CN and CZ1418LL, respectively (Fig. 3a,b).

A similar pattern was observed for plants grown in native soil, where MN1312CN and CZ1418LL exhibited a tendency for increase in H_2 production by 22% and 24% during the same time window, but these were not found to be statistically significant (Fig. 3c,d). However, the trends remained statistically significant when combining data from both soil types (Fig. 3e,f), despite apparent H_2 production being significantly lower in plants grown in native soil ($P < 0.0001$), likely due to soil-borne stressors or H_2 consumption by soil microbes.

Similar to nitrogen fixation, measurements of leaf TR (*E*) conducted concomitantly with H_2 sampling were found to significantly increase by 34% during the afternoon ($P = 0.01$) combining genotypes and soil types. Also in this case, only plants grown in sandy soil exhibited a significant increase in *E*, by 65% and 35% in the afternoon relative to the morning for genotypes MN1312CN and CZ1418LL, respectively (Fig. 4).

As could be seen on Fig. 4(c,d), measurements made on plants grown in native soil were much more scattered, resulting in non-significant differences in *E*. The remaining gas-exchange variables, namely g_s and *A*, did not vary significantly as a function of the time of the day for both soil types, with the only exception being a decrease in photosynthetic rate for genotype CZ1418LL during the afternoon ($P < 0.05$, not shown) in plants grown in native soil, but not in sand. Due to the scatter of the field data, there was no significant correlation between gas exchange of a plant and its H_2 production rate measured in the field, although in all cases, they positively co-trended (not shown).

Nitrogen fixation and gas-exchange response to VPD under controlled conditions

Because morning-to-afternoon increases in nitrogen fixation in the field could be potentially driven by changes in environmental

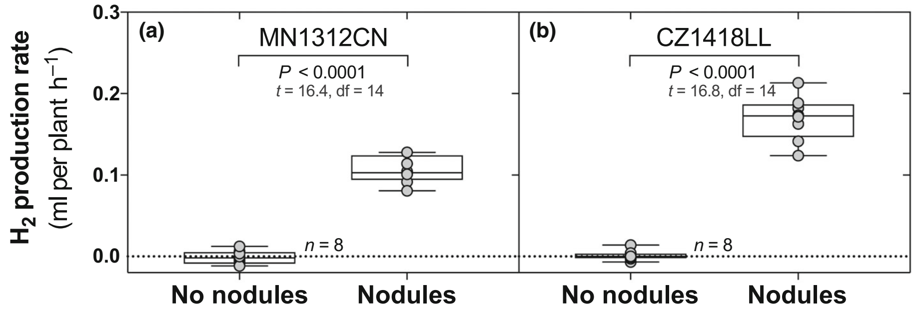

Fig. 1 Nitrogen fixation estimated via Hydrogen gas (H_2) production from nodulated and denodulated plants from soybean genotypes MN1312CN (a) and CZ1418LL (b). In the box-and-whiskers, the box represents the median and the 25th–75th percentiles, while the whiskers represent the largest and smallest values. *P*-values, *t*-statistic, and degrees of freedom shown are the result of *t*-tests between plants with no nodules and with nodules where *n* represents the total number of observations.

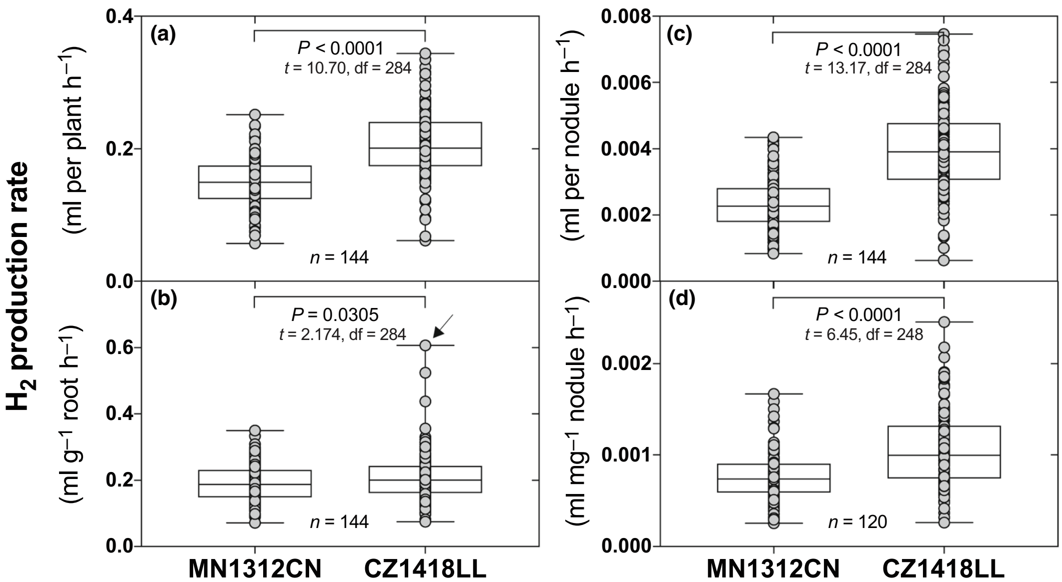

Fig. 2 Genotypic differences in soybean nitrogen fixation estimated via hydrogen gas (H_2) production by the root system of genotypes MN1312CN and CZ1418LL. H_2 production rates were expressed on a plant basis (a), normalized by root dry mass (b), nodule number (c), and nodule fresh mass (d). In (b), the arrow points to an outlier datum, the removal of which yields a nonsignificant difference between genotypes. In the box-and-whiskers, the box represents the median and the 25th–75th percentiles, while the whiskers represent the largest and smallest values. *P*-values, *t*-statistic, and degrees of freedom shown are the result of *t*-tests comparing genotypes where *n* represents the total number of observations. The relatively lower number of observations (*n*) in the case of (d) is the result of one group of plants overgrowing roots in the pouch, which obscured nodules in photographs.

conditions other than VPD, growth-chamber experiments were designed to examine VPD effect while keeping temperature, PAR, and air mixing constant in independent experimental runs (Table S3).

As shown in Fig. 5(b,d), H_2 production by nodules increased by 16% and 25% for MN1312CN and CZ1418LL, respectively, when VPD increased from 0.8 to 3 kPa (30°C held constant). However, a VPD increase in a lower range, from 0.5 to 2 kPa (25°C held constant), did not impact H_2 production (Fig. 5a,c). These same tendencies remained identical whether H_2 measurements were normalized based on root mass, nodule number, or nodule fresh mass (not shown). Regressing H_2 production against treatment VPD revealed a significant and positive slope for both genotypes (Fig. 5e,f), which remained significant when pooling phenotypic data of both genotypes (not shown, $P = 0.0008$).

Because the significant VPD effect on nitrogen fixation was only observed at 30°C, gas-exchange measurements were conducted concomitantly with H_2 measurements as VPD was increased from low (0.8 kPa) to high (3.0 kPa) while keeping this temperature and other environmental variables (PAR, air mixing) constant. This was done in order to test whether the relative change in leaf gas exchange as VPD was raised significantly increased H_2 production rate from the nodules combining data from both genotypes. These relative values were computed as the ratio of high-VPD to low-VPD measurements (i.e. fold increase). As reported in Fig. 6, among the examined gas-exchange variables, only the increase in TR was significantly and positively associated with an increase in H_2 production rate from the nodule as VPD was increased (Fig. 6a, $P = 0.043$). These correlations, however, were nonsignificant at the genotypic level

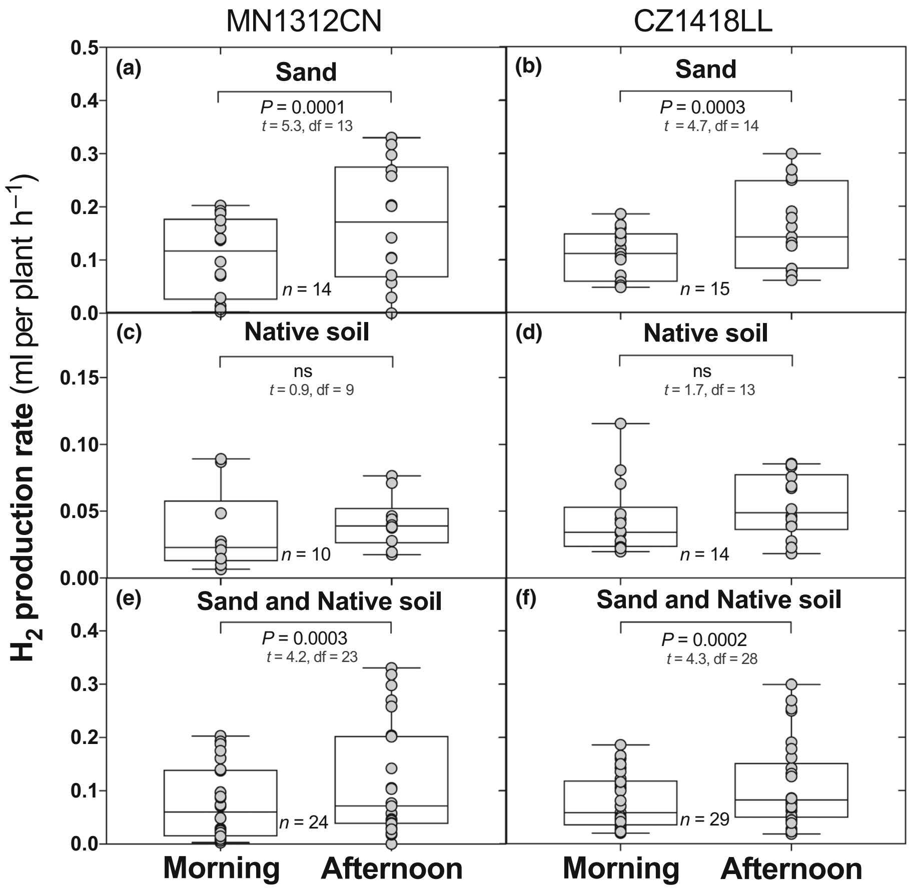

Fig. 3 Changes in nitrogen fixation estimated via Hydrogen gas (H_2) production by the root system as a function of the time of day under field conditions for soybean cultivars MN1312CN (a, c, e) and CZ1418LL (b, d, f). Nitrogen fixation was compared on plants grown in sand (a, b) and native soil (c, d), with (e, f) showing the results based on pooling data from both soil types. In the box-and-whiskers, the box represents the median and the 25th–75th percentiles, while the whiskers represent the largest and smallest values. *P*-values, *t*-statistic, and degrees of freedom are outputs of paired *t*-tests between morning (AM) and afternoon (PM) measurements conducted under conditions reported in Supporting Information Table S2. The letter *n* represents the total number of observations. Differences in the number of replications are a result of data logging errors where certain measurements were not properly recorded and the data were lost.

due to the large scatter of the data relative to the number of observations.

Discussion

Tracking root system H_2 production to nondestructively estimate and compare nitrogen fixation in legumes

Direct and real-time quantification of the rate of nitrogen fixation by legume root nodules in the field is a particularly tedious task and has been a long-standing challenge facing nitrogen fixation research. The hydrogen-production assay developed in this investigation, building on work of Witty & Minchin (1998), resolves major limitations typically associated with the more widely used ARA (Minchin *et al.*, 1983; Vessey, 1994; Smercina *et al.*, 2019). By measuring rates of hydrogen gas production by the nodules, a well-known by-product of nitrogen fixation (Witty, 1991; Hunt & Layzell, 1993; Oono & Denison, 2010), the method developed here avoids three drawbacks often associated with ARA. First, this method does not necessitate the complex logistics of using acetylene, a highly flammable gas, which is a source of safety concerns (Hunt & Layzell, 1993; Carson, 2002). Second, endogenous formation of ethylene gas can result in inaccurate estimation of nitrogen fixation as estimated using ARA (Smercina *et al.*, 2019). Third, this technique is less invasive than the use of saturating (10%) acetylene which can inhibit nitrogenase. Lower concentrations of acetylene can avoid this problem, but then, only a fraction of nitrogenase activity is measured.

Similarly, hydrogen production in air only measures a fraction of nitrogenase activity (Minchin *et al.*, 1983). The electron allocation coefficient (EAC) is the fraction of total nitrogenase activity going to nitrogen fixation rather than acetylene reduction or hydrogen production (Hunt & Layzell, 1993) and is complicated for acetylene reduction by diffusion issues (Denison *et al.*, 1983a, b). Actually measuring EAC in a particular context requires exposure to saturating acetylene or N_2-free $Ar:O_2$, respectively. Either can cause declines in nitrogenase activity (Minchin *et al.*, 1983), which may be avoided using short exposures. This is probably not possible in the field and was not done in the

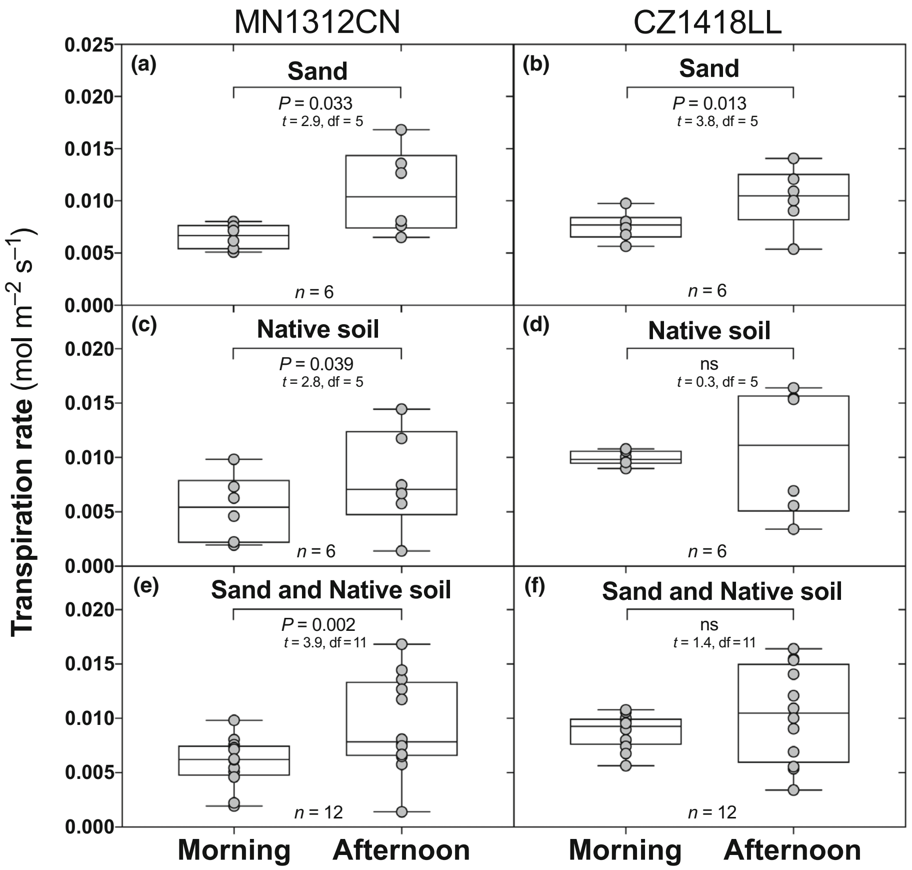

Fig. 4 Morning vs afternoon differences in transpiration rates measured on soybean plants grown in the field in sealed cylinders filled with sandy (a, b) or native soil (c, d) using a portable gas-exchange system, as a function of the genotype and combining data from both soil types (e, f). In the box-and-whiskers, the box represents the median and the 25th–75th percentiles, while the whiskers represent the largest and smallest values. *P*-values, *t*-statistic, and degrees of freedom shown are the result of *t*-tests morning and afternoon values where *n* represents the total number of observations.

laboratory experiments. Reliable published measurements of EAC mostly range from 0.6 to 0.7 (Hunt & Layzell, 1993), so assuming a value of 0.65 could result in errors up to 8% in the absolute rate of nitrogen fixation. If EAC of a given system is relatively stable; then, relative comparisons like ours should have even smaller errors. On bean plants, EAC changed only gradually over 6 wk (from 0.63 to 0.72), and there was no significant difference in EAC of two rhizobia strains that differed two-fold in NH_3-per-CO_2 efficiency (Oono *et al*., 2020). We are therefore confident that changes of 25–60% in hydrogen production (Figs 3, 5) with VPD correspond to real changes in nitrogen fixation and not to changes of a few percent in EAC.

The approach we used was reasonably successful in comparing relative rates of nitrogen fixation by soybean nodules. We did not detect H_2 production for non-nodulated plants while nodulated counterparts produced H_2 (Fig. 1), indicating that nodules are the primary source of the hydrogen gas sampled by the system. Additionally, consistent genotypic differences were observed in these responses (Figs 2, 3), which were maximized when normalizing H_2 production by nodule number (Fig. 2c), indicating that the H_2-sensing system is capturing a biological, genotype-dependent process, rather than a soil-based process *per se*. Furthermore, trends of root system H_2 production over time in the field were consistent over days, increasing in the afternoon independent of the soil medium, a finding that is consistent with early reports indicating that legume nitrogen fixation tends to increase in the afternoon (Johnson & Hume, 1972; Denison & Sinclair, 1985).

A significant limitation in the field is the potential presence of hydrogen-oxidizing bacteria in soils, which could result in weakening the 'signal' of H_2 production by the nodules (La Favre & Focht, 1983; Schuler & Conrad, 1991; Maimaiti *et al*., 2007). Additionally, the native field soil may have had nitrogen levels high enough to down-regulate atmospheric nitrogen fixation or pre-existing rhizobial populations which could have formed a symbiosis with the planted soybeans, possibly including strains expressing uptake hydrogenase (HUP^+) which recycle H_2 within the nodule (Evans *et al*., 1987; Godfroy & Drevon, 1991; Annan *et al*., 2012). Such H_2 signal weakening is supported by our field data, which shows that net hydrogen gas production was lower in plants grown in native soil, compared with those grown in columns filled with pure sand (Fig. 3), and in fact, the use of the sand as an alternative substrate in our experimental design was in anticipation of this possibility. The hypothesized presence of such bacteria in the native soil did not, however, generate an inconsistent trend in H_2 gas production relative to plants grown in the

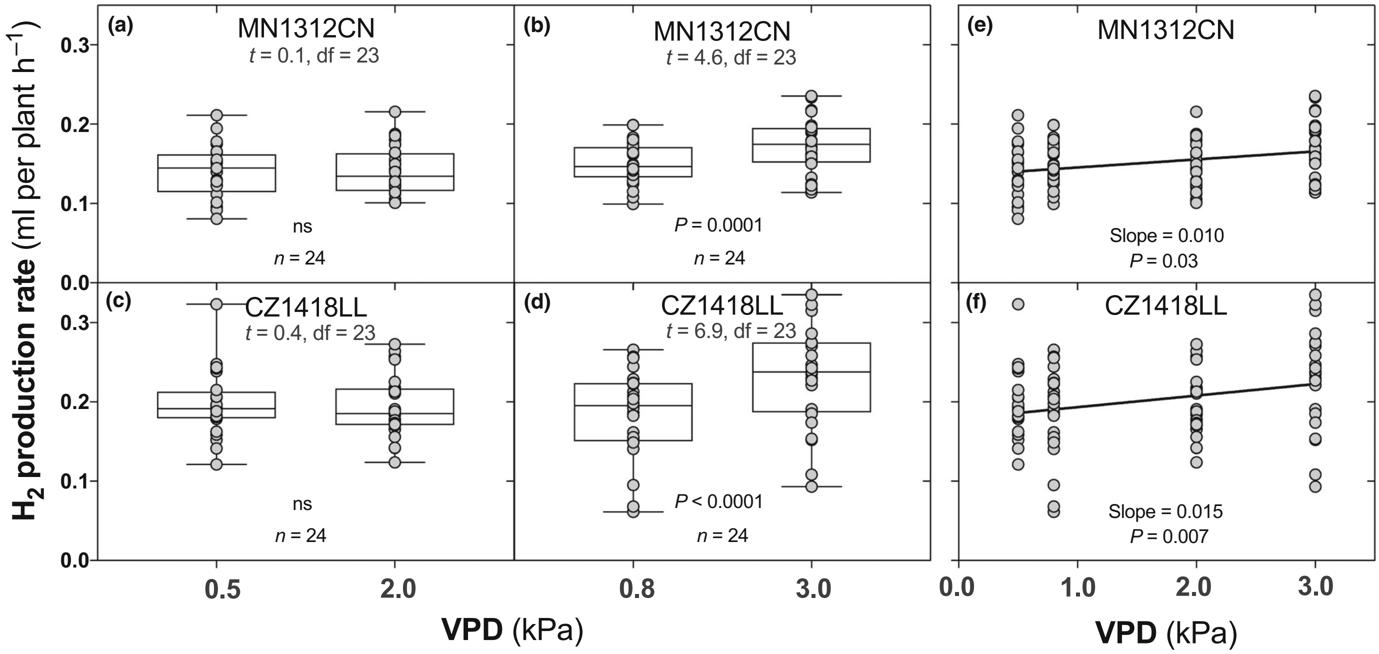

Fig. 5 Effect of vapor-pressure deficit (VPD) on nitrogen fixation estimated via H_2 production rate for soybean genotypes MN1312CN (a, b, e) and CZ1418LL (c, d, f). Nitrogen fixation was estimated by measuring hydrogen gas (H_2) production by the nodules under controlled-environment conditions (Supporting Information Table S3). Data are shown separately for the two temperature conditions (25°C, a and c; 30°C, b and d) and pooled as a function of VPD for regression analysis (e, f). In the box-and-whiskers (a–d), the box represents the median and the 25th–75th percentiles, while the whiskers represent the largest and smallest values. *P*-values, *t*-statistic, and degrees of freedom shown are the result of *t*-tests comparing treatments where *n* represents the total number of observations.

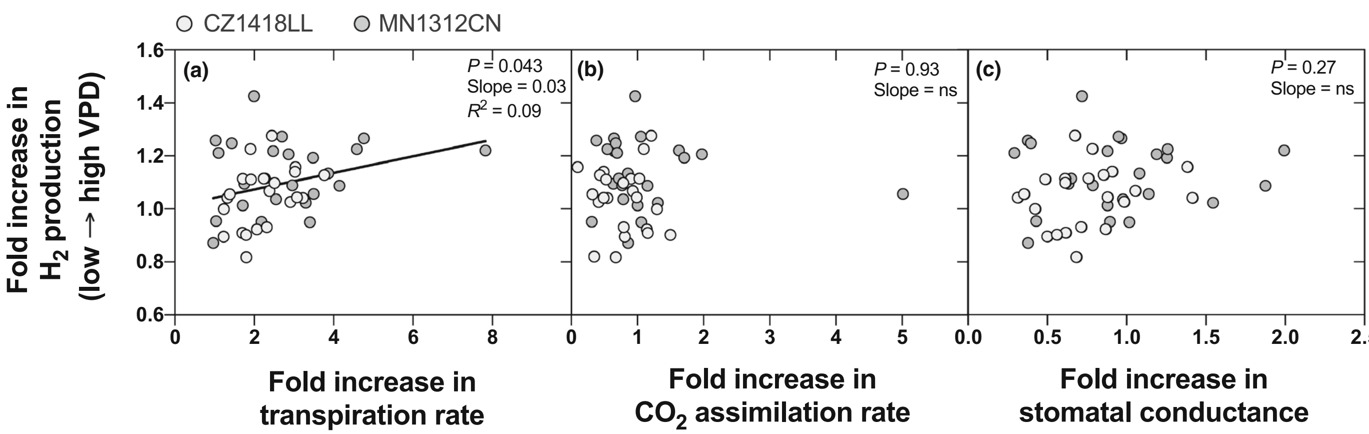

Fig. 6 Fold increase in nodule H_2 production rate in response to increase in leaf gas-exchange parameters (a, transpiration rate; b, photosynthetic assimilation rate; c, stomatal conductance) as vapor-pressure deficit (VPD) was increased from 0.8 to 3 kPa at a constant temperature of 30°C under well-watered conditions. Regression parameters (slope, R^2) are indicated only for the significant relationship. The two genotypes are represented by different symbols.

sand (compare (a), (b) and (c), (d) in Fig. 3). As a result, pooling nitrogen fixation data from both soil media still confirmed the earlier observations that nitrogen fixation increased in the afternoon (Johnson & Hume, 1972; Denison & Sinclair, 1985).

Nitrogen fixation increases in response to raising evaporative demand in soybean

A major finding of this work is the previously unreported result that, under well-watered conditions, nitrogen fixation in soybean, and presumably in other nitrogen-fixing plant species, increases as VPD was raised over a few hours, especially above 2 kPa (Fig. 5e,f). Consistent with our hypothesis, this increase in VPD led to rising nitrogen fixation by both soybean genotypes, by up to 25% (CZ1418LL) when measured at a constant temperature of 30°C. This increase was unlikely to be dependent on circadian effects or duration of the light period since independent experiments showed that H_2 production did not change as a function of the time of the day when VPD was not changed, in both genotypes and regardless of the temperature treatment (Fig. S4). If confirmed on a broader range of experimental conditions, genotypes, and species, this result points a much more important, and previously undocumented role of VPD not only on legume productivity but also on global nitrogen cycling.

A parsimonious explanation for the favorable VPD effect found in this study on well-watered plants would be its involvement in reducing nitrogen fixation feedback inhibition arising from the accumulation of ureides and amino acids in nodules, a regulation mechanism which has been extensively documented in soybean (e.g. Serraj *et al.*, 1999; Vadez & Sinclair, 2001; King & Purcell, 2005). This hypothesis is supported by the positive correlation observed between the relative increase in nitrogen fixation and TR as VPD was increased from 0.8 to 3 kPa in the controlled-environment experiment (30°C, Fig. 6a). This indicates that the increase in TR alleviated nitrogenase feedback inhibition either through the translocation of nitrogen compounds away from the nodules or by reducing lateral escape of ureides from xylem back to the phloem and then to the nodule (Parsons *et al.*, 1993; Bacanamwo & Harper, 1997; White, 2012). This potential link is further supported by a recent investigation by Pradhan *et al.* (2021) showing that xylem anatomical features that lead to sustained water transport are associated with alleviating soybean nitrogen fixation feedback inhibition and with findings that anti-transpirants reduced nitrogen fixation in pea (Aldasoro *et al.*, 2019). However, earlier results from de Silva *et al.* (1996) indicated that reducing TR by humidifying the air around the canopy did not significantly influence nitrogen fixation in soybean. Such seeming discrepancy from our results could be the result of several major methodological differences relative to our experimental setting. For instance, the high humidity treatment was imposed by sealing a mylar bag around the canopy, which might have interfered with temperature and PAR conditions. This treatment was maintained for 48 h and compared with a different set of control plants, while our treatment consisted of increasing VPD and measuring nitrogen fixation over a much shorter window (2–3 h) on the same plants.

The lack of correlation between H_2 increase and variation in A (Fig. 6b) indicates that no direct or immediate involvement of photoassimilate availability in nitrogen fixation response to VPD occurred in the timeframe of the controlled-environment study. Consistent with this, H_2 production in the field was not associated with variation in A, while g_{sw} remained unaffected and E increased (Fig. 4).

Overall, the results of this investigation point to a promoting role of VPD increase on nitrogen fixation in legumes under well-watered conditions, while showing evidence for genotypic variability in this response. Furthermore, this variability was identified within a narrow range of intensely selected elite germplasm, suggesting promising potential to breed for specific nitrogen fixation phenotypes in response to atmospheric drying. Future research is needed to confirm the feedback-inhibition hypothesis underlying this response and quantify the productivity benefits and tradeoffs associated with variation in nitrogen fixation response to raising VPD at the plant, ecosystem, and global scales.

Acknowledgements

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Competing interests

None declared.

Author contributions

DM contributed to the experimentation, data analysis, and writing. RFD contributed to the conceptualization and instrument design. WS contributed to the conceptualization, data analysis, and writing. All authors contributed to writing and agreed on the final version of the manuscript.

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Data availability

The data that support the findings of this study are available via the following Dryad link: 10.5061/dryad.h18931zrh.

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Daily time courses of environmental variables during nitrogen fixation measurements in the field.

Fig. S2 Diagram of the nitrogen fixation measurement setup deployed in the field and growth chamber.

Fig. S3 Time courses of nitrogen fixation rates during sampling in the field and growth-chamber environments.

Fig. S4 Lack of time-of-day effect on nodule nitrogen fixation.

Table S1 Average growth conditions during all experiments of the study.

Table S2 Environmental conditions during nitrogen fixation measurements in the field.

Table S3 Experiment designators and environmental conditions during the controlled-environment experiment.

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